THESIS SUMMARY

Surname, Name	FIKADU, Dagmawi Tarekegn
Student number (matricola)	VR509938
Degree course	Master's Degree in Molecular and Medical Biotechnology
Department	Department of Biotechnology, University of Verona
Thesis title	Curation and Cheminformatic Characterization of a Clinical-Stage GPCR Ligand Dataset for Machine-Learning-Based Binding Prediction
Principal Investigator	Prof. Alejandro Giorgetti
Supervisor	Filippo Baldessari
Academic year	2025–2026

BACKGROUND AND AIM

G protein-coupled receptors (GPCRs) are the largest family of membrane receptors in the human genome and the target of roughly a third of approved drugs. Predicting how tightly a small molecule binds a given receptor, from chemical structure alone, would allow large compound libraries to be prioritised computationally before any experiment is performed. The obstacle is not the modelling itself but the evaluation: chemical datasets contain dense series of close analogues, so a model tested on randomly selected compounds is largely re-predicting molecules it has effectively already seen, and the resulting performance is inflated. This thesis had two aims: to assemble and characterise a curated dataset of clinical-stage GPCR ligands, and to build binding-affinity models on that dataset under validation strict enough that the reported numbers can be trusted.

DATASET CURATION AND CHARACTERISATION

Human K_i binding data were retrieved from ChEMBL for compounds in clinical phases I–IV, restricted to molecules below 700 Da, and processed through a curation chain comprising salt removal, peptide exclusion, a flexibility filter and deduplication retaining the highest pK_i per molecule. The resulting set contains 3,011 compound–target records spanning 719 unique molecules, 47 targets and 18 receptor families, with a strong aminergic composition. Fifteen molecular descriptors were computed for every compound.

Two properties of the set proved consequential. First, selecting on clinical status rather than single-target potency yields extensively polypharmacological chemistry: molecules act on 4.19 distinct targets on average, and 64.7 % engage more than one. Second, ligand efficiency declines sharply with molecular size (Spearman $\rho = -0.77$): median efficiency falls from 0.52 to 0.24 between the smallest and largest compounds while median potency is essentially unchanged (6.66 against 6.51), so additional molecular mass purchases no additional affinity. Principal component analysis showed the descriptor space to be low-dimensional, with 65.5 % of variance in two components, and clinical phase did not separate meaningfully within it.

PREDICTIVE MODELLING AND VALIDATION

Two data-rich aminergic receptors, serotonin 5-HT_{2A} and dopamine D_3, were selected for modelling. The curated clinical set provides only 172 and 173 compounds per target, too few to train and validate honestly, so each was extended with preclinical ChEMBL compounds under identical curation rules, giving 4,620 and 5,488 molecules. Each molecule was represented as a 2,248-dimensional vector combining a 2,048-bit ECFP4 fingerprint with approximately 200 normalised physicochemical descriptors, and three tree ensembles were trained.

Validation used a repeated Bemis–Murcko scaffold split in which whole scaffold families are assigned to one side of the partition, so that every test compound is structurally novel to the model. Under this protocol the best model reached a mean held-out R^2 of 0.562 for 5-HT_{2A} and 0.644 for D_3 over twelve partitions, with root-mean-squared errors of 0.81 and 0.76 log units. Under conventional ten-fold cross-validation the same models score 0.679 and 0.717, and both estimates are reported so that the gap between them is explicit. The scaffold-split error corresponds to approximately 1.6 and 1.5 times the inter-laboratory reproducibility of pooled ChEMBL K_i measurements, indicating that a substantial share of the residual error originates in the labels rather than the model.

Recast as a virtual-screening classifier at a 100 nM potency cut-off, the models achieved Matthews correlation coefficients of 0.635 and 0.632, with balanced errors and calibrated output probabilities, and top-1 % enrichment at approximately 98 % of the ceiling imposed by the class balance. Four independent controls support these figures: a 100-permutation y-randomisation test ($p = 0.0099$), a nearest-neighbour read-across baseline that the models exceed by roughly 0.2 R^2, an ablation removing polypharmacological compounds, and a stricter similarity-cluster split. Under that last and most demanding partition performance approximately halves, and the model retains an advantage over the baseline for D_3 but not for 5-HT_{2A} — a limit that is reported explicitly. SHAP attribution indicates that the two receptors are governed by different property families, charge and electrostatics for 5-HT_{2A} and molecular topology for D_3.

CONCLUSIONS

The work delivers a curated and characterised clinical-stage GPCR ligand dataset, and affinity models whose performance is established under leakage-controlled evaluation and supported by four separate controls. The principal contribution is methodological: the reported figures are conservative estimates of prospective performance rather than the optimistic values that random-split evaluation produces, and the conditions under which the models fail are stated alongside those under which they succeed. Extension across the wider receptor panel, three-dimensional and structure-aware representations, and the inclusion of confirmed inactive compounds are the natural continuations.

Verona, 14 September 2026



Dagmawi Tarekegn Fikadu — VR509938

Signature of the candidate